

CS 564 Project Report

GenAI + Coding

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Executive summary

This report presents a single, holistic, exploratory case study of how EECS students at Oregon State University use generative AI (GenAI) in technical coding projects. We observed three students across three tasks of varying complexity — a full-stack Care Plan application (hard), a robotics arm assignment (medium), and a HackerRank Sudoku problem (easy) — and triangulated interview transcripts, observation notes, and prompt logs. Using a debugging codeset adapted from Lawrance et al. (2010) with two added codes (With AI, Information Gap), we find that students turn to GenAI primarily to fill information gaps, that AI reliance rises with task complexity (49.3% of session time on Care Plan vs. 6.4% on Robotics), and that working in a group lowers AI use. GenAI functioned as a scaffolding and gap-filling aid rather than a replacement for students' own problem-solving.

Research Questions and propositions	<u>Research question;</u> <u>How do EECS Students at OSU use GenAI for technical projects?</u> <u>Propositions:</u> 1) In what condition, would an EECS student tend to use GenAI? 2) When they use GenAI, what exactly did they do? 3) What portion of the time did they use GenAI?
Unit of analysis	<u>The EECS students at OSU is the unit of analysis.</u>
Type of case study	<u>It's a Single Holistic and Exploratory case study.</u>
Data collected	<u>Qualitative Data:</u> 1) <u>Observation:</u> 2) <u>Interview Transcripts:</u> 3) <u>Prompt Logs & Chat Histories:</u> 4) <u>Project Context Data:</u> 5) <u>Open-end Survey</u>

Design Type	<u>Single Holistic Case Study</u>
Code set origins	The study adapts Lawrance et al.'s debugging codeset to capture participant behavior in a natural field setting. To address generative AI use, two new codes were added: "With AI" (when participants used AI) and "Information GAP" (when AI was used to fill missing knowledge), with the latter informed by Subrahmaniyan et al.'s work and simplified to fit the RQ

A Case study of EECS students' use of GenAI in technical projects.

Introduction: (brief, usually 1 page)

- Background and motivation:

As GenAI development is growing, there is a need to monitor its use. Especially in academic settings using GenAI for some may even out the playing field allowing them to learn the material quickly while for others it may hamper their learning process. Hence it is important to assess how and why students use GenAI in technical projects.

Subjects of this case study are EECS majors who directly experience the phenomenon we are investigating. And since the subjects are friends of the researchers, they are more likely to provide honest feedback on sensitive topics.

- Research questions
 - How do EECS Students at OSU use GenAI for coding technical projects?
 - We aim to observe students as they use GenAI for both technical projects with a focus on using it to write code. We will keep student identities anonymous to handle privacy concerns and to encourage students to participate in learning outcomes.
 - Propositions:
 - In what condition, would an EECS student tend to use GenAI?
 - When they use GenAI, what exactly did they do?
 - What portion of the time did they use GenAI?
- Theory ties with prior work:

1. We are using an existing theory to assess **how** students use GenAI based on their coding behaviour. For this we are using prior work by Joseph Lawrance et al. who theorize that **human problem-solving is strongly shaped by cost-benefit information search strategies** rather than purely internal reasoning. We use the code debugging strategies shown below to study how students problem solve and at what point they use GenAI. To guide this assessment we will use the following categorization of usage:
 - a) Mapping mode: Does the student use GenAI to build a mental model of the problem
 - b) Observe-the-failure mode: Does the student use GenAI to be able to understand error messages
 - c) Locate-the-failure mode: Does the student use GenAI to find where the bug lies in their code
 - d) Fix-the-fault mode: Does the student use GenAI to fix faults in their code
 - e) Verify mode: Does the student use GenAI to verify correctness
2. Changzhao Wang et al. assess why students use GenAI for writing assignments. This is an exploratory study that investigates as part of its research questions what the strengths of AI were for the students for writing tasks. Their core finding was that **AI functions best as a drafting and scaffolding tool, not a replacement for human cognition**. This project will investigate whether this is valid in technical projects.
3. Subrahmaniyan et al study the adoption of software visualization tools and identified systematic obstacles during the “trial period”. Their core finding was that adoption of tools is not just about functionality and rather **depends on whether users can bridge information gaps quickly enough** to justify continued engagement. We study this in the context of technical projects and whether the theory stands for GenAI adoption as well.

Methods and case study design:

- Type of Case Study & Design Type
 - Single Holistic Case Study
 - This is a Single Case Study because we only have one case to study which is EECS students use GenAI for technical projects.
 - This is a Holistic Case Study because we only have one unit of analysis which is the EECS students at OSU.
- Research questions
 - How do EECS Students at OSU use GenAI for technical projects?
 - We aim to observe students as they use GenAI for technical projects with a focus on using it to write code. We will keep student identities anonymous

to handle privacy concerns and to encourage students to participate in learning outcomes.

- Propositions:
 - In what condition, would an EECS student tend to use GenAI?
 - When they use GenAI, what exactly did they do?
 - What portion of the time did they use GenAI?
- Tie each RQ/proposition to our data **and codes**

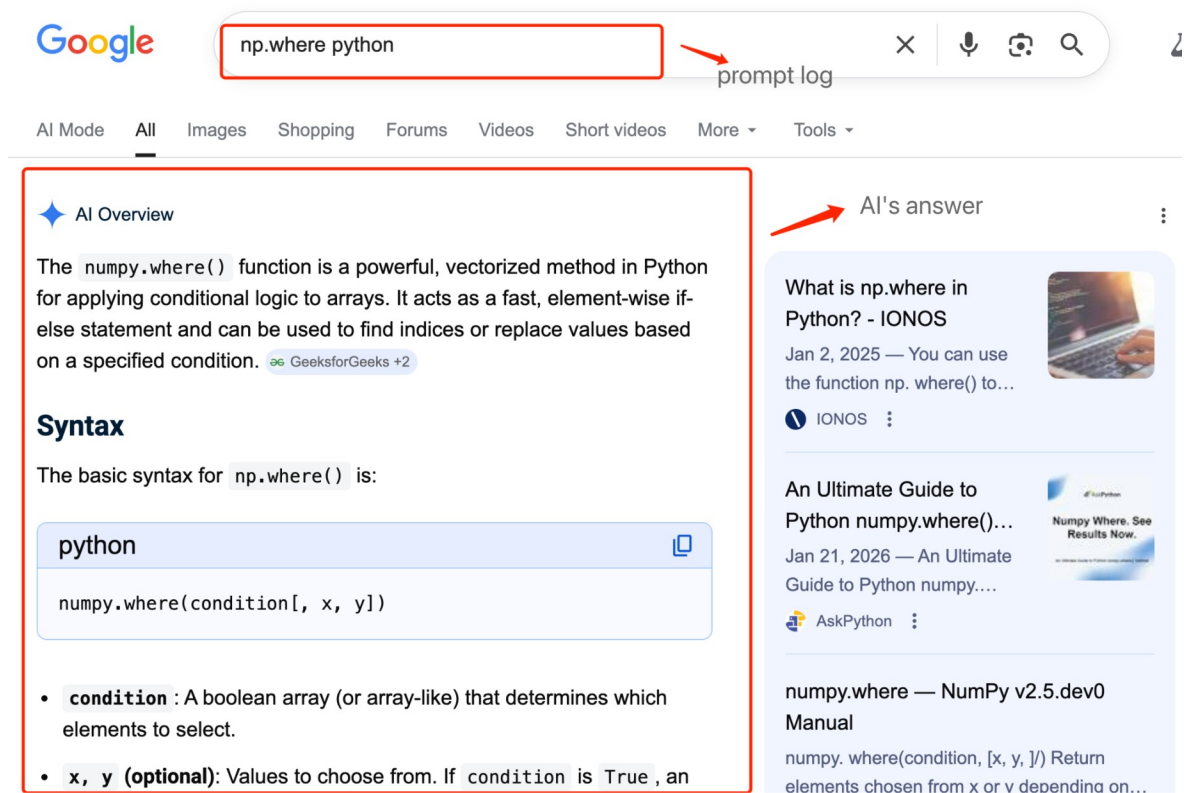
Propositions	How does the code/data answer each RQ/proposition?
How do EECS Students at OSU use GenAI for technical projects?	This research question will be answered by integrating the qualitative research results of each proposition. By analyzing the occurrence frequency, context and intent of the "With AI" label under different cognitive and behavioral states (such as "Mapping mode", "Verify mode" and "Fix the fault mode"), The aggregated data will show whether GenAI mainly serves as a brainstorming partner, code debugging assistant, or a direct code generator.
In what condition, would an EECS student tend to use GenAI?	This proposition will be answered by analyzing specific code tags that occur before the "With AI" tag or co-occur with it. By reviewing interview transcriptions and observation logs, look for moments such as "Information GAP" or "Observe and locate the failure mode" before using AI. The data will reveal the exact cognitive bottlenecks and technical triggering conditions that prompt students to turn to GenAI.
When they use GenAI, what exactly did they do?	This proposition will be answered by examining the specific code tags that occur after the "With AI" tag or co-occur with it. We will classify the AI interaction behaviors and observe whether students use them for high-level architecture planning (" Mapping mode "), verifying the solutions they propose (" Verify mode "), or seeking direct code corrections (" Fix the fault mode "). In addition, qualitative data from interview transcriptions (for example, specific Prompt words/prompts sent to the AI, and how the AI's output is integrated into the code) will provide conclusive evidence of their specific operational behaviors.
What portion of the time did they use GenAI?	This proposition will be answered by quantifying the qualitative observation data. By using the timestamps recorded side by side With the "With AI" tag in the observation log, we can calculate the exact duration of AI interaction events relative to the total time consumed by technical tasks. Furthermore, by analyzing the density and frequency of the "With AI" label throughout the dataset, a measurable proportion can be derived, thereby reflecting the overall time dependence of students on generative AI tools.

- Unit of Analysis
 - The EECS students at OSU is the unit of analysis.
 - We will monitor their use of GenAI as they work through **technical projects** and ask them to self-report their experience.
 - Wrt proposition 4 we determine code debugging strategies and will report what our observations are. To guide this assessment we will use the following categorization of usage (Joseph Lawrance et al. 2010):
 - Mapping mode: Does the student use GenAI to build a mental model of the problem
 - Observe-the-failure mode: Does the student use GenAI to be able to understand error messages
 - Locate-the-failure mode: Does the student use GenAI to find where the bug lies in their code
 - Fix-the-fault mode: Does the student use GenAI to fix faults in their code
 - Verify mode: Does the student use GenAI to verify correctness
 - Wrt. proposition 3 and 4, we will ask Open-ended questions.
 - Why did the student use GenAI? This question verifies the intent of the participant.
- Data collection
 - Observation: By observing the students regarding how they are using GenAI in their **technical projects**. Student identities such as name and university id have NOT been collected to keep privacy intact and encourage participation in the study.
 - Interview Transcripts: First, we got the permission from the participants. And the interview happened after the observation. The questions were mainly focused on what makes students make each movement, how they felt during the process, and their detailed experiences. This data was collected through unstructured interviews.
 - Prompt Logs & Chat Histories: The text prompts students typed into the GenAI tool and the code it outputted.
 - It will depend on the participants' comfort to reveal this data for us to add it to Verification of data we collected.
 - Help us explore "How do the students use GenAI for technical projects?"
 - Survey: We gave survey to all the participants using non-leading questions. We asked about their reasons for using or not using AI in 2-3 specific sentences. This data helps us understand their attitude towards AI and better answer "How do the students use GenAI for technical projects?"

- Data collection (For 3 project studies which includes Health Care Plan, Robotics Assignment & HackerRank Coding)
 - The field study was done on the Oregon State University campus. Locations included the common room of a robotics lab, student lounge in Kelly Engineering and study room 1161 at Valley Library. These places were chosen because we want participants to feel not too nervous, and more like they are coding in their normal and familiar environment
 - Participants and Roles:
 - In two interviews, we had 1 participant and 2 interviewers (1 and 2). However, in the third interview, there were 2 participants and the same 2 interviewers because it was a group assignment. This setup was kept during the whole interview. Having 2 interviewers was mainly for data triangulation and at the same time, it did not make the participants feel uncomfortable because of too many people.
 - Before the interview started, we got the participant's permission. We as interviewers told the participants that the interview was completely anonymous and it was only used for collecting data for our field study. This step was to protect participants' privacy and also reduce their worries, so they would not change their normal behavior when doing AI coding.
 - The transcript was turned on during the whole interview to make sure no conversation was missed. We used paper and pen to write down the participant's body language, actions like open tabs, and the time points. We also drew a diagram and took photos of the interview location.
 - Before starting, we as the interviewers told the participant not to feel nervous. Instead of getting right or wrong answers, we asked the participants to treat us as students, and teach us how to use LLM to help with their personal project, robotics assignment and coding task. This was to help participants feel more comfortable when explaining what they are doing, and also helped us collect more data for answering our research questions.
 - These interview questions mainly focused on
 - what the prompt was used for
 - what kind of answer the participant expected from the LLM
 - whether the LLM's answer met the participants' expectation after it was generated.
 - These interview questions were designed because we want to understand the motivation of EECS students when using GenAI for technical projects, and to get more details about this process.
 - All sessions involved participants working through technical problem-solving tasks while selectively using AI-based tools.

- Interview 1 (Care Plan): During the health care interview, the participant used LLM to build the minimum viable product (mvp) of the project. At the same time, the participant answered our questions with guidance and shared their thoughts during the process.
 - Interview 2 (Robotics): For the robotics assignment, two participants collaborated in a robotics lab, working with ROS topics, joint state controllers, gripper compatibility, and trajectory planning. Troubleshooting was primarily done through Google searches and occasional Google AI overviews, with minimal use of large language models due to their lack of context-specific accuracy.
 - Interview 3 (HackerRank): For the HackerRank Coding, one participant worked individually on a Sudoku board validation problem in Python, verbalizing their reasoning while debugging NumPy indexing issues. Google AI was mainly used for quick syntax reference, while LLMs were described as ineffective for debugging. Overall, the data showed that AI tools were treated as supplementary resources rather than primary problem-solving aids.
- After the interview, we also collected the prompt log from the interviewers and the project design documents as references. This helped us better understand the context, and to compare how participants deal with different types of coding tasks.
- Finally, we gave a survey to all the participants. We used non-leading questions so we wouldn't influence their answers. The survey asked two main questions about their experience with technical projects:
 - What are your reasons for using AI? We asked them to provide 2-3 complete and specific sentences to explain the situations or tasks where AI is helpful.
 - What are your reasons for NOT using it? We also asked them to write 2-3 complete and specific sentences to explain why they choose not to use AI in certain projects.
- Database structure
 - All raw data from Care Plans were translated from Chinese to English to ensure the consistency. Here's the link:
 - [original-design-document.md](#)
 - [original-transcript.docx](#)
 - [original-prompt-log.md](#)
 - Since the interviews were unstructured and Zoom failed to differentiate speakers, we manually separated the dialogues between interviewers and participants.
 - Field observations originally recorded by pen and paper, then were transcribed into English digital documents. Here's the link:

- [Hackerrank_Observation_Notes.pdf](#)
- [Robotics_Observation_Notes_1.jpeg](#)
- [Robotics_Observation_Notes_2.pdf](#)
- [CarePlan_Observation_Notes.pdf](#)
- For the prompt log, the Care Plan's prompt logs were in Markdown format, but we moved them to a spreadsheet. The data for HackerRank and Robotics was special because those students used Google Search and read the AI's answers. We only recorded the links at first. Then, we checked the links to find the search words and the AI's answers. We then put both into the spreadsheet as prompt logs and AI's answers to make all data look the same. Here's the link: [Prompt_Logs_Interview.docx](#)



- For the survey data, after looking at the first data, we saw that participants use AI very differently in different projects. To understand their attitude and more context, we sent a non-leading survey about why they use or not use AI. We pull all the answers in this Google Doc so we can analyze their differences later. Here's the link: [Survey Responses.docx](#)
- Finally, all processed data were segmented in the dataset according to Segmentation Rules mentioned below.

- Structure of the Spreadsheet:

- Here's the link:
1. [Dataset](#)
- The database includes the following components:

2. Interview Transcripts (3 total)

- Unstructured interview transcripts for each task (Care Plan, HackerRank, Robotics)
- Organized by segment number, speaker role (interviewer/participant), and spoken content

3. Interview Observation Notes (3 total)

- Researcher notes documenting participant behavior, pauses, confusion points, and interaction strategies during the coding process

4. AI Interaction Logs

- **Cursor AI prompt logs** capturing prompts used during code generation, debugging, and refinement
- **Google Search AI prompt logs** documenting external AI-assisted information-seeking behavior

5. Coding Process Annotations

- Transcript segments are coded using predefined categories such as:
 - With AI
 - Mapping Mode
 - Information Gap
 - Observe and locate the failure mode
 - Verify Mode
 - Fix the fault mode
- Description of the spreadsheet:
 - Each spreadsheet contains the following mandatory columns:

■ Segment	Num
Each interview transcript is divided into sequential segments (stanzas) based on shifts in conversational context, task focus, or coding activity. A segment represents a coherent unit of interaction that can be independently analyzed and coded.	
■ Person	
This column identifies the speaker associated with each segment. Entries indicate whether the content was produced by the <i>interviewer</i> or the <i>participant</i> , enabling analysis of interaction dynamics and role-specific contributions.	
■ Content	
This column contains the verbatim textual data for each segment, including	

spoken dialogue, explanations, and task-related reasoning expressed during the interview. The content serves as the primary qualitative data source for coding and thematic analysis.

- Segmentation

- Main data comes from the voice records of interview participants, the observation of participants, and the prompt logs they provided. To make it easy to build our dataset, we transcribe the voice records into text, and combine with the observer's notes and prompt logs to build our final dataset.
- Not only that, to make sure the information is complete and easy to understand, in transcripts, we use one meaningful question from interviewer as the segmentation; in observation notes, we follow each behavior of participants that we recorded; in prompt logs, we put each participant's question to AI and the AI's answer as one segment.
- Besides that, for the Interview 2 (The robotics assignment) the participant was working on a group assignment. The team contained 2 people. the role of the apprentice was lower/contained. This was done to observe back and forth discussion between the team and the apprentice intervened to ask AI specific questions when we saw them using GenAI. For this situation we segmented the data by participant 1, participant 2 and interviewer segments

- Coding process

- The Codeset is adapted from J. Lawrance et al.'s work “How Programmers Debug, Revisited: An Information Foraging Theory Perspective”. Taking into account that this is a field study we ensured not to push the candidates to use AI and observe them in their neutral settings of solving assignments. We needed a code set that can be used to judge the participants overall behaviour for which the debugging modes worked well. We added two new codes as per our requirement around GenAI. First, we added “With AI” which serves as a direct signal for us to study under what circumstances the participant has actively used AI. Secondly we added “Information GAP” which is a common use case of AI in debugging which was not covered by the paper. We refer to Subrahmaniyan et al.’s work “Software Visualization for End-User Programmers: Trial Period Obstacles” to understand what Information gaps are but merged them into a single code since it matched our RQ to understand how EECS students use AI in technical projects.
- Code set: We divided the code set amongst pairs of members of the team. Each pair coded 24/204 segments randomly selected from the dataset. Our first Jaccard test revealed that Information Gap, Observe/Locate failure mode, and With AI codes were replicable with a score 100%, 75% and 85% respectively. We adjusted the definition of Mapping mode, Verify mode and fix the fault mode to be replicable

and used a new set of segments where 100%, 100% and 83% jaccard score was observed. The final code set is shown below:

Code	Definition	Example
With AI	Participant is actively using GenAI	“For this one, I copy specific customer requirements to AI and let AI generate it.”
Mapping mode	The participant is building a mental model of what the nature of the whole problem is and understanding the big picture.	“Let me first go through everything and understand the full problem before trying to solve it.”
Information GAP	It is regarding the specific approach/question/method in order to find the correct solution which the participant is trying to figure out.	“I need more specific details about the approach before I can figure out the correct solution.”
Observe and locate the failure mode	Participant is debugging and is trying to locate the bug by understanding error messages. They don't know where the problem lies but are aware of what the problem is.	“The error is pointing to this line, so the bug might be coming from here.”
Verify mode	Participant is facing a specific problem and is proposing a specific solution to fix it. They are trying to understand if the solution will work or not .	“I believe this solves it, but I need to test and confirm.”
Fix-the-fault mode	The participant is fixing a fault in their current implementation. The participant already knows what the specific problem/bug is and is making changes .	“The problem was in this loop condition. I'll modify it.”

Results

In this section we will discuss observed trends in the data after our code set was applied and answer the RQ: How do EECS Students at OSU use GenAI for technical projects?

Tasks

The three interviews conducted included three very different project types. In order to better

interpret the results it is useful to consider the task difficulty of the projects. We took into consideration whether the project had a single objective or multiple objectives, whether the participant was starting from scratch or had pre-existing code to build on.

Interview 1 (*Care Plan*): This was a multi-objective problem where the participant had to build multiple apis. The participant had to start from scratch and write a lot of code like docker files and apis. For those reasons this was a *hard difficulty problem*. In this case we observe that they need help from AI in order to generate the minimal viable product like the sample project template so that it will be faster for them in order to make changes on top of that.

Interview 2 (Robotics): This was a single objective problem of recording movement in an arm. The added difficulty was due to the participants having to understand a large code base and manufacturing documents to understand. For those reasons this was a *medium difficulty problem*. In this case the participant used Google AI to search how specific robot parts operated.

Interview 3(HackerRank): This was a single objective problem where participant had to create a sudoku solver. The participant had to start from scratch and write a single method. For those reasons this was an *easy difficulty problem*. In this case the participant primarily used Google AI to see examples of how packaged methods were used.

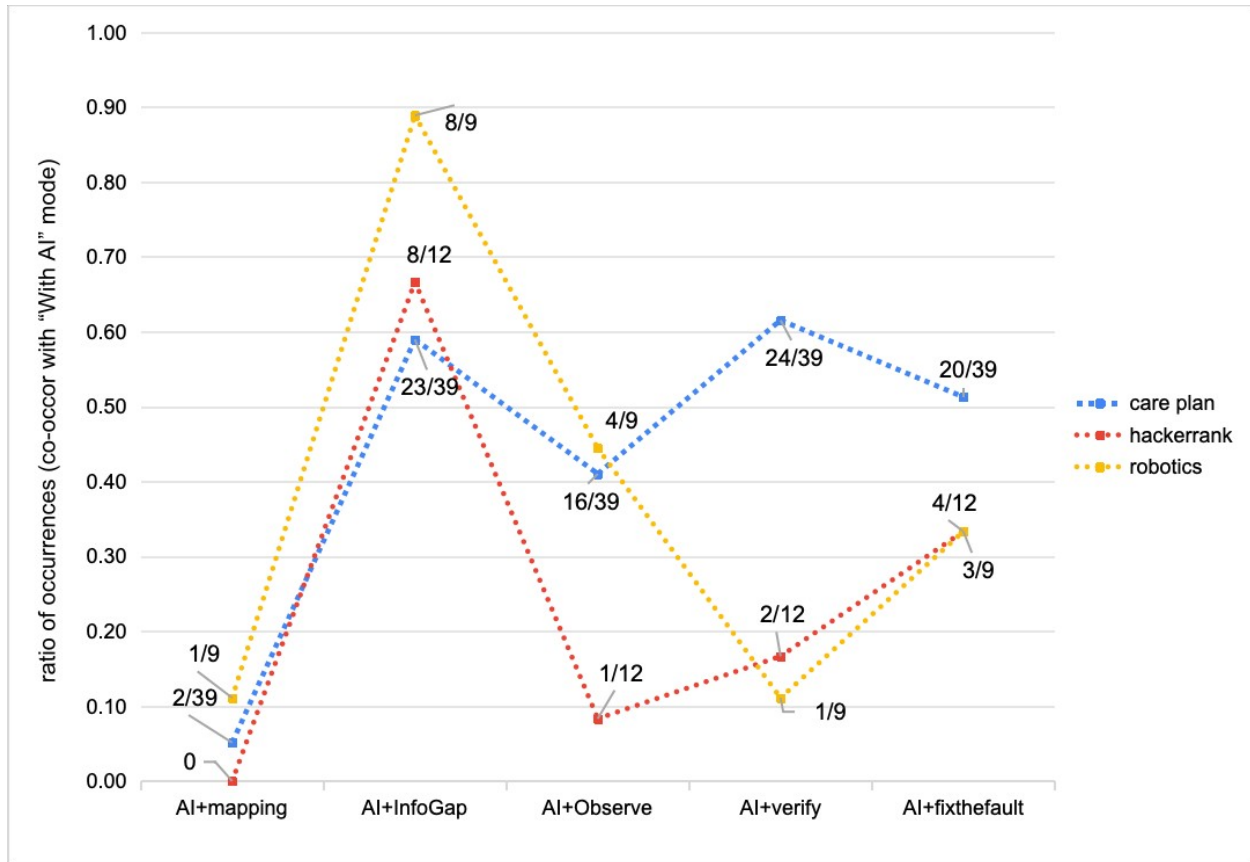


Figure 1: Co-occurrence distribution of "With AI" label and specific cognitive or debugging behavior patterns in different interviews.

This line plot shows the distribution of students' specific intentions and behaviour in deciding to use GenAI when the three students were handling Care Plan, HackerRank tests, and Robotics respectively. The x axis represents the co-occurrence of the five behavioral codes (Mapping mode, Information Gap, Observe and Locate the failure mode, Verify mode, and Fix the fault mode) with the "With AI" label; the y axis represents the proportion of that co-occurrence frequency over all "With AI" segments in the entire interview. The numerical value next to the data point indicates the specific occurrence rate (number of specific co-occurring segments / total number of segments labeled "With AI" in this task). The line plot indicates that, although three participants had very different specific tasks, filling the information gap (AI+InfoGap) was the primary common reason that led them to use GenAI.

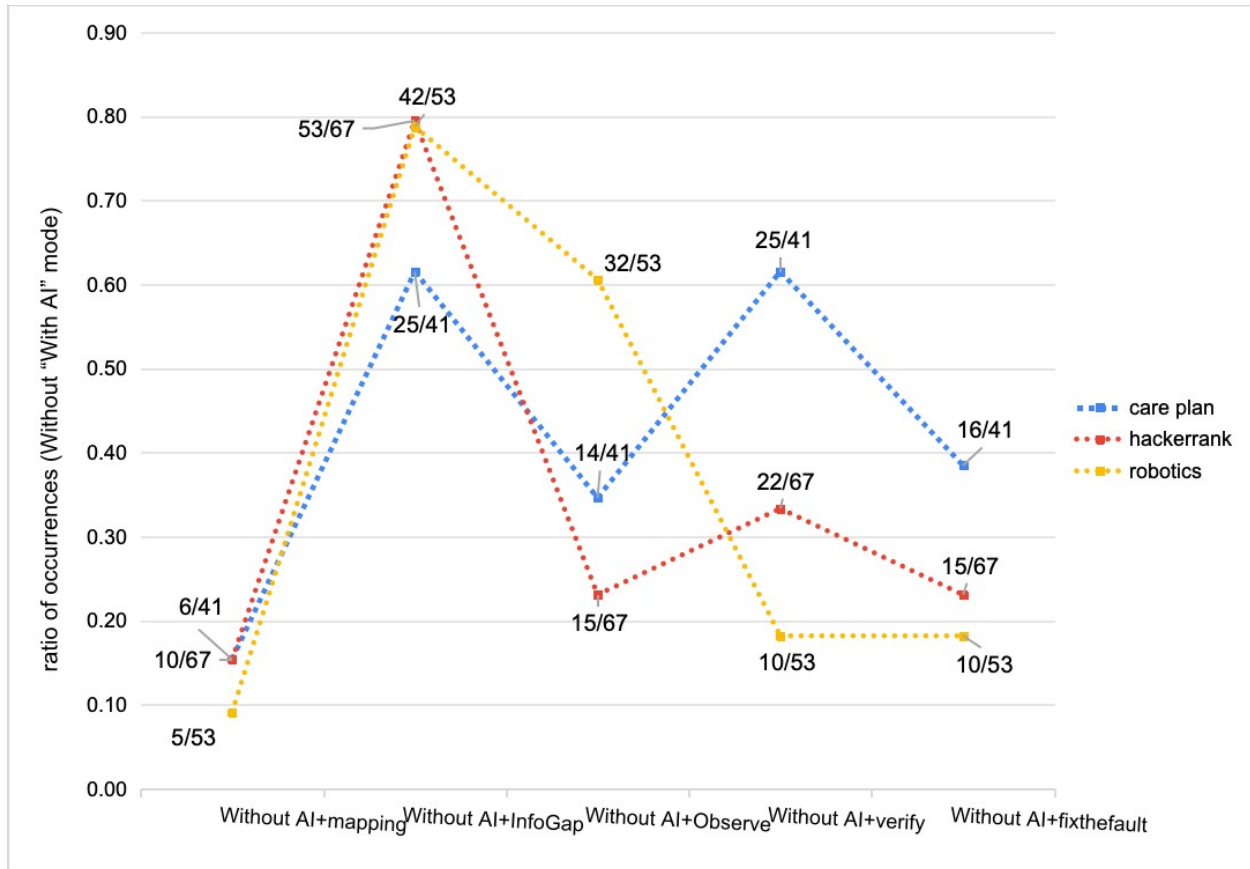


Figure 2: Occurrence distribution of specific cognitive or debugging behavior patterns that without "with AI" label in different interviews.

This line plot shows the distribution of students' specific intentions and behaviour in deciding to **not** use GenAI when the three students were handling Care Plan, HackerRank tests, and Robotics respectively. The x axis represents the occurrence of the five behavioral codes (Mapping mode, Information Gap, Observe and Locate the failure mode, Verify mode, and Fix the fault mode) **without** the "With AI" label; the y axis represents the proportion of that occurrence frequency over all the segments without "With AI" label in the entire interview. The numerical value next to the data point indicates the specific occurrence rate (number of specific behaviour **does not co-occurring with AI** segments / total number of segments **without** "With AI" label in this task). The line plot indicates that, although three participants had very different specific tasks, filling the information gap (**Without AI+InfoGap**) was the primary common behaviour and for "Without AI+Observe" and "Without AI+verify" three participants showed entirely different behaviour.

Proposition 1: In what condition, would an EECS student tend to use GenAI?

To measure the condition of the participant and gauge their behaviour we considered two data segments before each occurrence where the participant started using AI so as to find out what could have caused the participant to use AI. This establishes prior behaviour. We also looked at

two data segments after the participant has finished using AI to track the impact of AI use. We show the triangulated results below:

Interview Id			Mapping mode	Information Gap	Observe and locate the failure mode	Verify mode	Fix-the-fault mode
1	Care Plan	Interview	3	12	9	14	10
		Observation	2	9	3	4	3
		Prompt Log	1	9	8	10	8
2	Robotics	Interview	0	2	3	0	1
		Observation	0	3	3	1	1
		Prompt Log	0	2	0	0	1
3	HackerRank	Interview	0	2	1	4	1
		Observation	0	4	1	1	1
		Prompt Log	0	2	0	0	2

Table 1: Before Participant used AI what did they do

Interview Id			Mapping mode	Information GAP	Observe and locate the failure mode	Verify mode	Fix-the-fault mode
1	Care Plan	Interview	3	10	7	15	10
		Observation	1	6	3	3	3
		Prompt Log	1	9	8	11	8
2	Robotics	Interview	0	2	1	0	1
		Observation	0	4	1	1	0

		Prompt Log	1	2	0	0	1
3	HackerRank	Interview	0	3	2	2	2
		Observation	0	2	1	1	1
		Prompt Log	0	2	0	0	2

Table 2: After Participant used AI what did they do

We observe that In Interview 1 (Care Plan project) the participant had varied behaviour across all modes of operation and is observed in all three data sources. This suggests that the candidate preferred using AI frequently. More importantly before using AI as well as after AI use the participant is primarily in verify mode as indicated by the interview transcripts and prompt logs and observation notes, followed by Information Gap. This suggests that the participant is actively verifying the deployment here.

For Interview 2 (Robotics project) we observe that the primary behaviour before and after AI use is the Information gap mode. This suggests the participants are still searching for the solution and as mentioned before has to understand a lot of code before they can solve the problem.

Participant in Interview 3 (HackerRank) is in both fix the fault mode and Information gap mode. We observed the participant actively problem solving and showing a good logical understanding of the problem.

Proposition 2 : When Participant/Participants used AI what did they do?

Care

Plan:

In the Care Plan task, participant used AI mainly to identify information gaps, verify their ideas, and fix faults in their implementation. The way he used AI depended on how complex the task was. In many cases, the participant first used AI to validate his idea before implementing it. He wanted to check whether their approach would work or not. According to our codeset, this behavior falls under the Verify Mode. As shown in the table, AI was used mostly for verification in the Care Plan task, and after that it was used to fix bugs or issues in the implementation. For example, participant mentioned:

"Actually, I just want to let AI help me clear my mind. My core goal today is to implement an MVP. For this care plan project, I am thinking of using two APIs: one POST and one GET. POST will upload user data from the nurse, and GET will use that data to generate a prompt for the LLM and return the result to the nurse. I want to tell Cursor this idea first to see if this process will work. So the LLM is mainly helping me organize my thoughts and check if my idea is correct." This shows that the participant was using AI to organize their thinking and confirm whether the

approach was correct, which corresponds to the Verify Mode in our codeset. After this stage, Information Gap and Observe and Locate Failure Mode also appeared. In these situations, the participant used AI to find missing information related to the task and to identify bugs or errors when the code did not run correctly.

Hacker

Rank:

In the HackerRank task, the participant mainly used AI to fill information gaps related to the coding task. Most of the time, AI was used to check syntax and library usage, especially for pandas and numpy. The participant used AI to quickly understand how to write certain commands or functions. Compared to the Care Plan task, AI was used less frequently for debugging or fixing errors. This pattern is visible in the Interview, Observation, and Prompt Log data, where most AI usage was related to finding missing information rather than fixing problems. For eg: Participant tries to search for the following in Google Gemini regarding numpy library: “The numpy.where() function is a powerful and efficient method in Python for applying vectorized conditional logic to arrays. It is used for conditional selection, value replacement, and finding indices of elements that meet specific criteria.

Syntax

The basic syntax is as follows, where parameters x and y are optional:
Python

```
numpy.where(condition, [x, y, ]/)
```

condition: A boolean array or array-like expression.
x, y (optional): Values from which to choose. The shapes of x, y, and condition must be broadcastable.”

Robotics: In the Robotics task, the participants also used AI mainly to find missing information related to the project. However, the use of AI for locating failures, verifying solutions, or fixing issues was relatively lower compared to the other tasks. This suggests that AI was mostly used as a support tool for understanding concepts or retrieving information, rather than for debugging or validating implementations.

For example, we can see from the observation notes in one instance: “*they used Google and they are trying to get a solution.*” In another instance, participants searched for technical information related to the robot hardware, such as “*Google Kinova Kortex*” and “*Google Robotiq URDF.*” Participants also looked up resources related to performing the task itself, such as “*Kinova arm pick and place task.*”

These examples show that participants used AI and online resources mainly to understand the robotics framework, find references, and guide their implementation while working on the task.

Interview Id			Mapping mode	Information GAP	Observe and locate the failure mode	Verify mode	Fix-the-fault mode
1	Care Plan	Interview	1	7	6	10	8
		Observation	0	7	2	3	4
		Prompt Log	1	9	8	11	8
2	Robotics	Interview	0	1	1	0	1
		Observation	0	4	3	1	1
		Prompt Log	1	3	0	0	1
3	HackerRank	Interview	0	2	0	1	0
		Observation	0	3	1	1	1
		Prompt Log	0	3	0	0	3

Table 3: When Participant used AI what did they do

Proposition 3: What portion of the time did they use GenAI?

As Table 4 shows, the proportion of time spent using GenAI varied considerably by task type. Care Plan Interview had the highest AI usage at 49.30% of total session time, followed by HackerRank at 17.75% and robotics at the lowest with 6.39%.

Interview	Total Time	Time with AI	Time without AI	AI / Total Ratio
Interview 1 – Care Plan	1:15:00	0:37:00	0:38:00	49.30%
Interview 2 – Robotics	0:39:54	0:02:33	0:37:21	6.39%
Interview 3 – HackerRank	0:43:24	0:07:42	0:35:42	17.75%

Table 4: How much time was spent using AI

This pattern is further supported by Figure 3 This bar plot shows the occurrences ratio of "with AI" or "without AI" behaviour when the three students were handling Care Plan, HackerRank tests, and Robotics respectively. The red bar represents the occurrences ratio of the segments with "with AI" labels. The blue bar represents the occurrences ratio of the segments without "with AI" labels. The numerical value next to the data point indicates the specific occurrence rate (number of specific behaviour segments / total number of all the meaningful segments which has at least one label). Where Interview#1 (Care Plan) again showed the highest with AI ratio at 39/65 segments, compared to Interview # 3 (HackerRank) at 12/51 and Interview#2 (Robotics) at 9/42.

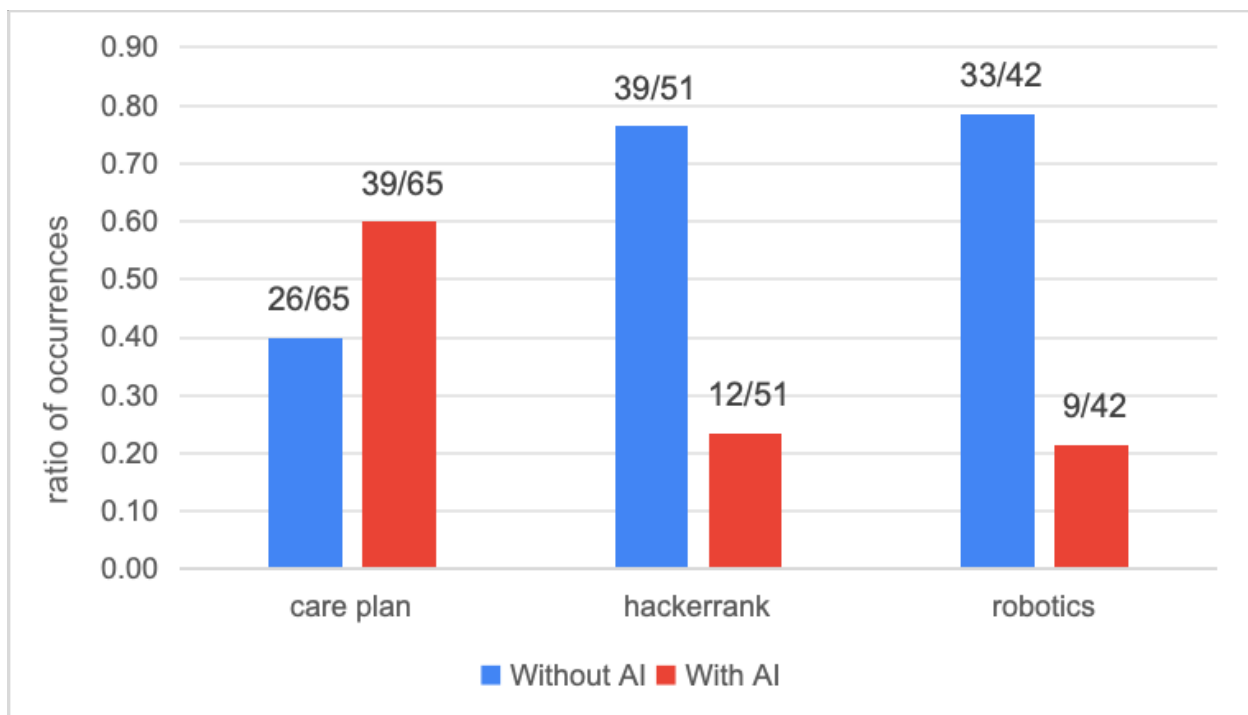


Figure 3: Occurrence distribution of with and without "With AI" label in different interviews.

Why are the AI usage times so different?

Analysis 1: Participant attitudes impacted AI usage time.

From the above discussion we can conclude that there are behavioural differences in the participants who used AI a lot and those that rarely used it. Participant who used AI for the majority portion of the time were observed in the verify mode right before and after AI use. To further

investigate why this phenomenon occurs we asked participants the following two questions regarding their reasons for using or not using AI:

1. What are your reasons for using AI?
2. What are your reasons for NOT using it?

Interview 1 (Care Plan): Positive
"I use AI because I believe mastering these tools is a vital skill in today's world. It significantly boosts my coding efficiency by breaking down complex problems into clear, manageable tasks. This way, I can "learn by doing" while I work, making the whole process much more productive."

"I don't use AI on the rare occasions when its answers don't meet my expectations; in those cases, I prefer to find the answer myself from other sources. Also, when I really want to learn a new concept from scratch and get hands-on experience, I choose to work without AI to ensure I truly understand it."

As we can see, the participant preferred the learning by doing approach and viewed AI as a must have skill.

Interview 2 (Robotics): Neutral

"I generally use AI for cases I know are more general, for example, getting the arm to do a pick and place task. Pick and place is a very common experiment setup and I know there could be multiple answers/approaches to it."

"The times I won't use AI is when what I'm searching for is novel, for example, certain bugs that come up when working with the arm and are unique to it. Documentation or public forums are better for these cases"

As we can see, the participant viewed AI as a conglomeration tool for general problems and prefers documentation for niche problems. The participant is very nonchalant about their opinion of AI.

Interview 3 (HackerRank): Negative

"I use AI as a way to summarize material from multiple sources for quick viewing to find an initial starting point for the decision/task I need to do."

"I don't use AI to do my work because I want to gain the experience in actually creating/doing the task so I can learn. By having an AI do it instead, I don't gain the experience as fast as I would without."

We can also see that the participant displayed a strong desire for self learning and also used AI as a conglomeration tool. However, here they used it to get a starting point for a task. The participant views AI negatively as a way to take away learning from them.

Normally, we would expect the neutral attitude participant (Robotics) will use AI more than the negative attitude participant (HackerRank). However, the Robotics AI usage (6.39%) was actually much lower than that of HackerRank (17.75%). This leads to our analysis 2&3 in which we try to find the deeper reasons rather than the attitude towards AI.

Analysis 2: Project type impacted AI usage time.

As we mentioned above, three participants were working on three totally different tasks. For careplan, it's a multi-objective and full-stack task and it has only involved one participant. For HackerRank, it's a single-objective and single method task which also have involved one participant. But for robotics, it's a single-objective task with a large code base and manufacturing documents, and it actually involved two participants. So this shows that the different AI usage time is not only related to the participants' attitude, but also relevant to the property of each specific task. In addition, there's a difference between three projects which is the number of participants involved in the project. This leads to our analysis 3.

Analysis 3: Individual vs. group work impacted AI usage time.

For the robotics project, two participants were working on it together. Having a partner who could discuss the ideas naturally results in decrease of the usage of AI. It's also corresponding with the lowest time usage of AI.

In conclusion, positive attitudes can increase AI use in multi-objective and complex projects. But for the teamwork project with the large code base and manufacturing documents, those properties overpowered the participants' personal attitudes, resulting in the lowest AI usage time in the Robotics project despite their neutral attitude.

Triangulation:

Triangulation was used during The Interviews:

During the field interview we collected what the participant said (Interview transcripts), what they typed into the AI (Prompt logs) and what we observed them doing (observation notes) detailed in the data collection section. Each of these data sources help us triangulate our conclusions and provide reliability of the behaviour of the participants. In our study, triangulation meant that we did not depend on just one source to understand what the participant was doing or thinking. If all three sources showed the same behavior, we could draw meaningful conclusions. If the sources did not match, that usually meant something more interesting was happening and we needed to look deeper.

The Three Sources We Triangulated:

1. Interview Transcript (What the participant said)
During each session, the interviewer asked questions such as “Can you explain the goal of this prompt?” or “Did the AI’s answer meet your expectations this time?”

These questions were meant to help participants explain what they were thinking at that moment. The interviewer listened carefully to understand their reasoning process, for example, whether they were trying to understand the problem, checking if the solution was correct, or struggling because they were missing some information.

2. Observation Notes (what we saw them do): While one team member was engaging the participant with questions, another was watching and writing on paper independently. We recorded timestamps, body language, screen activity, and behavioral cues separately from what was being said. For example:

"17:15 Gets an error. Suspects Docker is not installed on the laptop." our observer logged this before the participant verbally confirmed it in the transcript

"They are on the tip of the chair leaned forward. They are trying their best." (HackerRank), physical engagement that confirmed that they are trying to solve the problem by themselves and not relying on AI to fix the issue. These notes were written without reference to what was being transcribed, making them a truly independent data point.

3. Prompt Logs (what they actually typed): After each session, we collected what participants actually sent to the AI. This was the most honest record, it revealed intent without requiring participants to explain themselves. For instance, the Care Plan participant verbally said they were "just trying to get the process work," but their prompt log showed:

"I need to generate a minimum viable product based on this requirements document. My current idea is to use 2 APIs... Do you think any other APIs are needed?"

Triangulation for Proposition 1&2:

Table 5 shows our triangulation from three different data sources: Transcript, Observation logs and Prompt logs for three different participants. This following table indicates the co-occurrence of specific behavioral code and the “with AI” code, and each checkmark indicates one segment with those specific codes.

		With AI + Mapping Mode			With AI + InfoGap			With AI + Observe		
		Transcript	Observation logs	Prompt logs	Transcript	Observation logs	Prompt logs	Transcript	Observation logs	Prompt logs
Proposition 1 & Proposition 2: Co-occurrence triangulation in different interviews, different data source	Careplan	✓		✓	✓	✓	✓	✓	✓	✓
	Robotics			✓	✓	✓	✓	✓	✓	✓
	Hackerank				✓	✓	✓		✓	
		With AI + Verify			With AI + Fixthefault					
		Transcript	Observation logs	Prompt logs	Transcript	Observation logs	Prompt logs			
		✓		✓	✓	✓	✓			
	Careplan	✓	✓	✓	✓	✓	✓			
	Robotics		✓		✓	✓	✓			
	Hackerank	✓	✓			✓	✓			

Table 5: Triangulation table for proposition 1&2. ✓ = the co-occurrence of one code (Mapping / Info Gap / Observe & Locate / Verify / Fix-the-Fault) and the “with AI” code as present in that source for these propositions. Blank = no evidence coded.

Threats to validity:

1. Replication Validity:

- a. In this study we did not verify how much experience the user had with AI in the past.
- b. In this study we did not verify how much prior experience the participants had with the task itself. This can affect how much AI is used. The expectation here would be that the AI use will be for writing purposes not coding purposes.

2. Internal Validity:

- a. The Interviews were conducted by team members separately in different settings.
- b. The Robotics project was a pair project and the presence of a human teammate to discuss ideas may have indirectly reduced the use of AI

3. How survey get participant attitude: Since we do not have code sets in order to map the survey responses to quantitative measures of attitude, surveys pose a threat to validity. The responses from the survey are not verified against any code sets and is an intuitive generalized view of the participants' responses to “Why they use AI” and “Why they don't use AI”.

4. Continuing issue for before/after segments: We observe that the behaviour of the participants remain the same before AI use, during AI use and after AI use as shown in Table #1-3. This can happen as each before and after segment is tagged relative to individual segments that use AI. However, most of the times when participants use AI they do so in continuous segments. This causes overall in the segments considered for analysis and causes the three tables to show similar results.

5. Construct Validity: from Fig1 & 2 from results, Further Analysis showed that the participants behaviour was task dependent rather than AI dependent. We should have measured the difference of behaviour with AI usage and without AI usage which is most likely to change some outcomes of our field study

Conclusion:

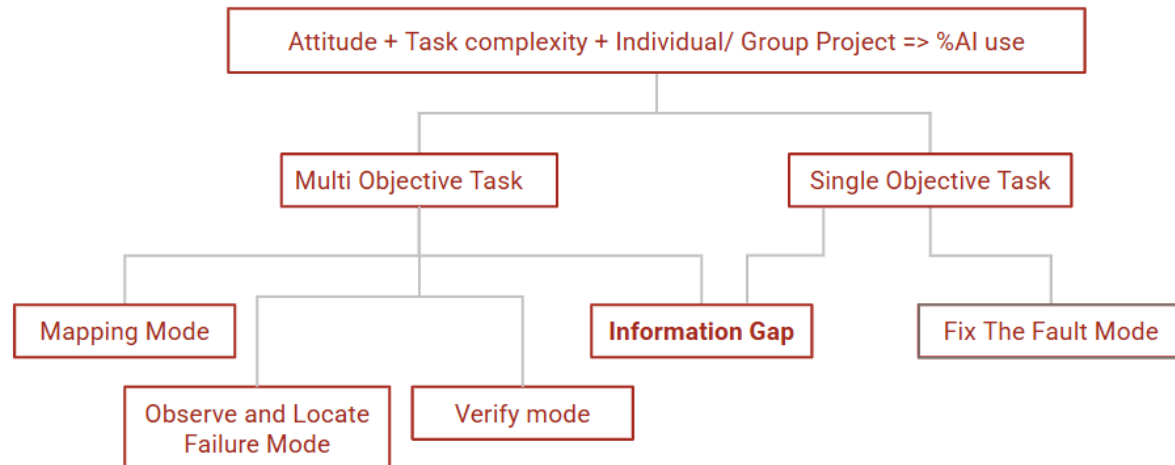
Through this work we have tackled an important question during the rise of GenAI use: “How and Why students use GenAI in technical projects”

The case studied here are EECS major students at Oregon State University. We interviewed three students and observed their natural problem solving behaviours when dealing with technical projects. The phenomenon we are studying is students use GenAI during these tasks.

We collect data from three sources and analyse the data using codes that help us track student behaviours during the coding and development process. The analysis answers the RQ: How do EECS Students at OSU use GenAI for coding technical projects?

Our primary contributions and findings are listed below:

1. The primary usage of AI was to fill an Information Gap. Our results align with Subrahmaniyan et al. as such that Information GAP is the primary use case of AI adoption. For the Interview #1 (Care Plan) had the highest use of AI in verify mode, information gap and fix the fault mode in that order. This is shown through Figure #1 and Table #3. Which answers proposition 2: When they use GenAI, what exactly did they do?
2. Results are triangulated across data sources as shown in Table #1-3. Exceptions include observe and locate failure and verify mode in robotics where the prompt logs did not show this.
3. The more complicated the task the more AI dependence was observed. As shown in Table 4 and Figure 3, the highest AI use was observed in the Care Plan project which answers proposition 3: What portion of the time did they use GenAI?
4. Through data triangulation, we observe that Interview #1 uses AI heavily and in varied contexts as seen in Table #1 and 2, while in Interview #2 and #3 Low AI usage is correlated with using it during Information Gap and Verification stages. This answers proposition 1: In what condition, would an EECS student tend to use GenAI?



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4. Joseph Lawrance et al. (2010). How programmers debug, revisited: An information foraging theory perspective | IEEE Journals & Magazine | IEEE Xplore.
<https://ieeexplore.ieee.org/document/5674060/>

Appendix

Dataset:

<https://docs.google.com/spreadsheets/d/1Yrgo5su-wNf3aMO7l8qe1RUv27qPILozENDpkvXHoeY/edit?gid=0#gid=0>

Interview guide:

<https://docs.google.com/spreadsheets/d/1uRLl3EgAzb6aCesBQhknqUFcP5CZuCMkOlxDt0Kl96w/edit?usp=sharing>

Google Drive:

https://drive.google.com/drive/folders/1vPnZeMGbKSj2_xpk7on75NWEHAuVxLUL?usp=sharing